

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$+C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$
								$-\frac{1}{2} C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger$		$-\frac{1}{2} C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger$
								$C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger$		$C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger$
								$-\frac{\overline{h y_{1,i_1,r_1}} y_{1,i_2,r_2} S_{1,j_1} \overline{S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}}{\epsilon}$		$-\frac{\overline{h y_{2,i_1,r_1}} y_{2,i_2,r_2} S_{2,j_1} \overline{S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}}{\epsilon}$
cont.								$+ \overline{h y_{1,i_1,r_1}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} S_{1,j_1}^\dagger S_{1,j_2}^\dagger$		$+ \overline{h y_{2,i_1,r_1}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger$
cont.								$+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}$
								$\overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}$		$\overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}$
								$-\frac{\overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}}{\epsilon}$		$-\frac{\overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}}{\epsilon}$
								$-\frac{1}{2} \overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}$		$-\frac{1}{2} \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$+C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$
								$-\frac{1}{2} \sqrt{3} C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger$		$-\frac{1}{2} \sqrt{3} C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger$
								$\sqrt{3} C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger$		$\sqrt{3} C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger$
								$-\frac{5 \overline{h y_{1,i_1,r_1}} y_{1,i_2,r_2} S_{1,j_1} \overline{S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}}{3\epsilon}$		$-\frac{5 \overline{h y_{2,i_1,r_1}} y_{2,i_2,r_2} S_{2,j_1} \overline{S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}}{3\epsilon}$
cont.								$+\frac{1}{3} \overline{h y_{1,i_1,r_1}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} S_{1,j_1}^\dagger S_{1,j_2}^\dagger$		$+\frac{1}{3} \overline{h y_{2,i_1,r_1}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger$
cont.								$+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}$
								$\sqrt{3} \overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}$		$\sqrt{3} \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}$
								$-\frac{\sqrt{3} \overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}}{\epsilon}$		$-\frac{\sqrt{3} \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}}{\epsilon}$
								$-\frac{1}{2} \sqrt{3} \overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}$		$-\frac{1}{2} \sqrt{3} \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2 \overline{h y_{1,i_1,r_1}} y_{1,i_2,r_2} S_1^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_1^\dagger \right)^2}{\epsilon}$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	$+\frac{2 \overline{h y_{2,i_1,r_1}} y_{2,i_2,r_2} S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2}{3\epsilon}$
								$+C_{S_1^\dagger S_1} S_1 S_1^\dagger$		$+C_{S_2^\dagger S_2} S_2 S_2^\dagger$
								$-C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_1 y_{1,i_1,r_1} S_1^\dagger$		$-\frac{C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_2 y_{2,i_1,r_1} S_2^\dagger}{\sqrt{3}}$
								$2 C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_1 y_{1,i_1,r_1} S_1^\dagger$		$-\frac{2 C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_2 y_{2,i_1,r_1} S_2^\dagger}{\sqrt{3}\epsilon}$
cont.								$+\frac{1}{2} S_1^2 \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$-\frac{4 \overline{h y_{2,i_1,r_1}} y_{2,i_2,r_2} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
cont.								$2 \overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}$
								$-\frac{\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}}{\epsilon}$		$-\frac{2 \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$
								$-\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$-\frac{\overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}}$
								$-\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$-\frac{\overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}}$
cont.								$-\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$-\frac{\overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}}$
cont.								$-\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(2)}$
								$-\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$2 \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(2)}$
								$-\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$-\frac{2 \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$
								$-\overline{h y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$		$-\frac{\overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} \lambda_{S_2}^{(2)}}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}\right)^\dagger^2}{3\epsilon}$ $+ C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$ $- \frac{C_{S_1^\dagger S_1} \hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$ $- \frac{2C_{S_1^\dagger S_1} \hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$+ D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$- \frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}\right)^\dagger^2}{\epsilon}$ $+ C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$ $- C_{S_2^\dagger S_2} \hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}y_{2,i_1,r_1}S_{2,j_1}^\dagger$ $- \frac{2C_{S_2^\dagger S_2} \hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$- \frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+ \frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^1 \lambda_{S_1}^{(1)}$ $- \frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^1 \lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$		$+ \frac{1}{2}S_{2,j_3}^2\left(S_{2,j_1}\right)^\dagger^2 \lambda_{S_2}^{(1)}$ $- \frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}\right)^\dagger^2 \lambda_{S_2}^{(1)}}{\epsilon}$
								$- \frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^1 \lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+ \frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^2 \lambda_{S_1}^{(2)}$ $- \frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^2 \lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$		$- \hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}\right)^\dagger^2 \lambda_{S_2}^{(1)}$
								$- \frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^2 \lambda_{S_1}^{(2)}}{\sqrt{3}}$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}\right)^\dagger^2}{3\epsilon}$ $+ C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$ $- \frac{C_{S_1^\dagger S_1} \hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$ $- \frac{2C_{S_1^\dagger S_1} \hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$+ D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$+ \frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}\right)^\dagger^2}{3\epsilon}$ $+ C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$ $- \frac{C_{S_2^\dagger S_2} \hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}}$ $- \frac{2C_{S_2^\dagger S_2} \hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
								$- \frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+ \frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^1 \lambda_{S_1}^{(1)}$ $- \frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^1 \lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$		$- \frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+ \frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}^1 \lambda_{S_2}^{(1)}$ $- \frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}^1 \lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$
								$- \frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^1 \lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+ \frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^2 \lambda_{S_1}^{(2)}$ $- \frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^2 \lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$		$- \frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}^1 \lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+ \frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}^2 \lambda_{S_2}^{(2)}$ $- \frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}^2 \lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$
								$- \frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}^2 \lambda_{S_1}^{(2)}}{\sqrt{3}}$		$- \frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}^2 \lambda_{S_2}^{(2)}}{\sqrt{3}}$

Field combinations 2 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{\epsilon}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}$	$(S_2)^\dagger D_{\mu_1}(S_2)$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$

Field combinations 3 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$ $+\frac{2h\overline{y_{1,i_1,r_1}}cg_{j_2,j_3}^1S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{1,i_1,r_1}}cg_{j_2,j_3}^1S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^2\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^2\lambda_{12}^{(2)}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^2\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^2\lambda_{12}^{(2)}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(cg)_{j_2,j_3}S_{2,j_1}^\dagger\lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(cg)_{j_2,j_3}S_{2,j_1}^\dagger\lambda_{S_2}^{(1)}$ $+\frac{h\overline{y_{1,i_1,r_1}}cg_{j_1,j_3}^1S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{1,i_1,r_1}}cg_{j_1,j_3}^1S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0			$-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}}CPR_{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$		$\frac{2\sqrt{3}h\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_{T3}\text{Bar}(cg)_{a_1,j_1,a_2,j_2}}{\epsilon}$ $-\sqrt{3}h\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_{T3}\text{Bar}(cg)_{a_1,j_1,a_2,j_2}$ $-\frac{4h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$+\text{CNewScalar1NewScalar2}hS_{1,j_1}S_{2,j_2}\text{Bar}(cg^3)_{j_1,j_2}$ $+\frac{1}{2}C_{S_2}^\dagger S_2h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg^3)_{j_1,j_2}$ $+C_{S_2}^\dagger S_2h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg^3)_{j_1,j_2}$
cont.										$-\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}}CPR_{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$ $-\frac{2\sqrt{3}h\overline{y_{1,i_1,r_1}}\lambda_{T3}cg_{a_1,j_1,a_2,j_2}^1y_{2,i_1,r_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$ $-\sqrt{3}h\overline{y_{1,i_1,r_1}}\lambda_{T3}cg_{a_1,j_1,a_2,j_2}^1y_{2,i_1,r_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger S_{1,j_2}^\dagger$ $+\frac{2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(1)}$	$+\frac{5h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(cg^3)_{j_1,j_2}\text{Bar}(cg^3)_{j_3,j_4}}{9\epsilon}$ $+\frac{1}{9}h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(cg^3)_{j_1,j_2}\text{Bar}(cg^3)_{j_3,j_4}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(cg^3)_{j_2,j_3}S_{1,j_1}^\dagger}{3\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(cg^3)_{j_2,j_3}S_{1,j_1}^\dagger}{3\sqrt{3}}$ $+\frac{10h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_2}\text{Bar}(cg^3)_{j_2,j_3}S_{2,j_1}^\dagger}{3\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_2}\text{Bar}(cg^3)_{j_2,j_3}S_{2,j_1}^\dagger}{3\sqrt{3}}$		
cont.										$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2\lambda_{H1}^{(2)}$		$+\text{HCNewScalar1NewScalar2}cg^3_{j_1,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{1}{2}C_{S_2}^\dagger S_2h\overline{y_{1,i_1,r_1}}cg^3_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+C_{S_2}^\dagger S_2h\overline{y_{1,i_1,r_1}}cg^3_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
cont.										$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2\lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2\lambda_{H1}^{(2)}$		$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $-\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$	
cont.													$-\frac{10}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{2}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{8h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
cont.													$-\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{8}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{2}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$
cont.													$+\frac{10h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}cg^3_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}cg^3_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}cg^3_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}\epsilon}$
cont.													$-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}cg^3_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$\begin{aligned} & -\frac{5h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\text{cg}^3_{j_1,j_3}\text{c}^{\text{E}3}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger S_{2,j_4}{}^\dagger}{9\epsilon} \\ & +\frac{1}{9}h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\text{c}^{\text{E}3}_{j_1,j_3}\text{c}^{\text{E}3}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger S_{2,j_4}{}^\dagger \\ & \quad +S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)} \\ & -\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon} \\ & -\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)} \\ & -\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon} \\ & -\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)} \\ & +\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}^3)_{j_2,j_3}S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & +\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}^3)_{j_2,j_3}S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}} \\ & \quad -\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)} \\ & \quad +\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)} \\ & +\frac{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}}{\epsilon} \\ & +\frac{3}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)} \\ & -\frac{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}}{\epsilon} \\ & -\frac{3}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)} \\ & +\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}}{\epsilon} \\ & +\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)} \\ & -\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}}{\epsilon} \\ & -\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)} \\ & +\frac{2h\overline{y_{1,i_1,r_1}}\text{cg}^3_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & +\frac{h\overline{y_{1,i_1,r_1}}\text{cg}^3_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}} \\ & \quad +S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)} \\ & -\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon} \\ & -\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)} \\ & -\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon} \\ & -\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)} \\ & +\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}^3)_{j_2,j_3}S_{2,j_1}{}^\dagger \lambda_{S_2}^{(1)}}{\epsilon} \\ & +\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}^3)_{j_2,j_3}S_{2,j_1}{}^\dagger \lambda_{S_2}^{(1)} \\ & +\frac{h\overline{y_{1,i_1,r_1}}\text{cg}^3_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger S_{2,j_3}{}^\dagger \lambda_{S_2}^{(1)}}{\epsilon} \\ & +\frac{1}{2}h\overline{y_{1,i_1,r_1}}\text{cg}^3_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger S_{2,j_3}{}^\dagger \lambda_{S_2}^{(1)}}{\epsilon} \end{aligned}$
													$\begin{aligned} & +\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F} \\ & +\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F} \\ & +\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F} \\ & -\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger}{\epsilon} \\ & -2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger \\ & \quad +H_{a_2}S_1H_{a_1}{}^\dagger S_1{}^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1\lambda_{H1}^{(1)} \\ & -\frac{2h\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1\lambda_{H1}^{(1)}}{\epsilon} \\ & -h\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1\lambda_{H1}^{(1)} \\ & -\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_2j_1y_{1,i_2,r_1}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon} \\ & +\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_2j_1y_{1,i_2,r_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
										$-\frac{2h\overline{y_{2,i_1,r_1}}S_1S_2j_2y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{h\overline{y_{2,i_1,r_1}}S_1S_2j_2y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}CPR\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$	$-\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+S_2S_{1,j_2}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$
cont.								$+\frac{2h\lambda_{T3}L_{a_3,i_1}CPR\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$ $-\frac{2h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{2h\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$ $-\frac{2h\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.									$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$ $-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1}CPR\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$	$-\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$ $-\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $-\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $+\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
cont.								$-2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1}CPR\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$ M_F	$-2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$ $-\frac{2h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $-\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}$
cont.									$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$ $-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$

[illegible]

Field combinations 4 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	$S_1 \ell$	$S_2 H$
cont.									$-\frac{\sqrt{3} h \overline{y}_{1,i_3,r_1} \overline{l}_{a_1,i_1}}{4 M_F} C P_R \overline{l}_{a_2,i_2} {}^T c g_{a_1,j_1,a_2,j_2}^{2, S_1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{1,i_3,r_2}$	$-\frac{1}{2} \sqrt{3 h \overline{y}_{2,i_1,r_1}} H_{a_2 S_2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} {}^2 \lambda_{H2}^{(2)}$
									$-\frac{3 \sqrt{3} h \overline{y}_{1,i_3,r_1} \overline{l}_{a_1,i_1}}{8 M_F} C P_R \overline{l}_{a_2,i_2} {}^T c g_{a_1,j_1,a_2,j_2}^{2, S_1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{1,i_3,r_2}$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{2,i_3,r_1} \overline{l}_{a_1,i_1}}{4 M_F} C P_R \overline{l}_{a_2,i_2} {}^T c g_{a_1,j_1,a_2,j_2}^{2, S_1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_1}$	
									$3 \sqrt{3} h \overline{y}_{2,i_3,r_1} \overline{l}_{a_1,i_1} C P_R \overline{l}_{a_2,i_2} {}^T c g_{a_1,j_1,a_2,j_2}^{2, S_1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_1}$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{2,i_3,r_1} \overline{l}_{a_1,i_1}}{c M_F} C P_R \overline{l}_{a_2,i_2} {}^T c g_{a_1,j_1,a_2,j_2}^{2, S_1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2}$	
									$-\frac{\sqrt{3} h \overline{y}_{2,i_3,r_1} \overline{l}_{a_1,i_1}}{c M_F} C P_R \overline{l}_{a_2,i_2} {}^T c g_{a_1,j_1,a_2,j_2}^{2, S_1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2}$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{1,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} 3 \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{1,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
cont.									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$	
									$-\frac{2 M_F}{4 c M_F} \sqrt{3} h \overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{1,i_3,r_3} \overline{l}_{a_2,i_2} {}^T C P_L L_{a_1,i_1} y_{2,i_3,r_2} \text{Bar}(e g_2)_{a_1,j_1,a_2,j$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_1	ℓ	S_2	H
$cont.$													$-\frac{\overline{hy_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger T_{H^2,a_1,a_2,j_2,j_1}}{\sqrt{3}}{}^2\lambda_{H^2}^{(2)}$

Field combinations 5 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F$	$S_2 \ell$	$F B_{\mu\nu}$
									$\begin{aligned} & + \frac{\overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{2, i_1, r_1} y_{2, i_2, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger}{2M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{2, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger}{2\epsilon M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{2, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger}{2\epsilon M_F} \\ & - \frac{\sqrt{3} h \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_2} S_{2, j_1}^\dagger S_{2, j_2}^\dagger}{4\epsilon M_F} \\ & - \frac{3\sqrt{3} h \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_2} S_{2, j_1}^\dagger S_{2, j_2}^\dagger}{8M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{2, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_2} S_{2, j_1}^\dagger S_{2, j_2}^\dagger}{4\epsilon M_F} \\ & - \frac{3\sqrt{3} h \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{2, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_2} S_{2, j_1}^\dagger S_{2, j_2}^\dagger}{8M_F} \\ & - \frac{h \overline{y_{2, i_1, r_1}} \overline{y_{2, i_2, r_1}} \overline{L_{a_2, i_2}} \epsilon^T C P_L L_{a_1, i_1} S_{2, j_1} S_{2, j_2} \text{Bar}(\epsilon \mathbb{E} 1)_{a_1, j_1, a_2, j_2} \lambda_{S_2}^{(1)}}{2\epsilon M_F} \\ & - \frac{h \overline{y_{2, i_1, r_1}} \overline{y_{2, i_2, r_1}} \overline{L_{a_2, i_2}} \epsilon^T C P_L L_{a_1, i_1} S_{2, j_1} S_{2, j_2} \text{Bar}(\epsilon \mathbb{E} 1)_{a_1, j_1, a_2, j_2} \lambda_{S_2}^{(1)}}{2M_F} \\ & - \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{2, i_1, r_1} y_{2, i_2, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{S_2}^{(1)}}{2\epsilon M_F} \\ & - \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \epsilon \mathbb{E} 1_{a_1, j_1, a_2, j_2} y_{2, i_1, r_1} y_{2, i_2, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{S_2}^{(1)}}{2M_F} \end{aligned}$	
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cont.										
cont.										
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & + \frac{2}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} y_{2, i_2, r_2}} S_{2, j_2}^2 y_{2, i_1, r_2} y_{2, i_2, r_1} \left(S_{2, j_1}^\dagger \right)^2 \\ & - \frac{2C_{S_2^1 S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_{2, j_1} y_{2, i_1, r_1} S_{2, j_1}^\dagger}{\sqrt{3}} \\ & - \frac{4}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} y_{2, i_2, r_2}} S_{2, j_1} S_{2, j_2} y_{2, i_1, r_2} y_{2, i_2, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger \\ & - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_{2, j_3} S_{2, j_4} y_{2, i_1, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger I_{S_2, j_3, j_1, j_4, j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}} \\ & - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_{2, j_3} S_{2, j_4} y_{2, i_1, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger I_{S_2, j_3, j_1, j_4, j_2} \lambda_{S_2}^{(2)}}{\sqrt{3}} \end{aligned}$	$-\frac{1}{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\begin{aligned} & -2C_{S_2^1 S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_2 y_{2, i_1, r_1} S_2^\dagger \\ & -2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_2^2 y_{2, i_1, r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)} \end{aligned}$	$-\frac{1}{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & + \frac{2}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} y_{2, i_2, r_2}} S_{2, j_2}^2 y_{2, i_1, r_2} y_{2, i_2, r_1} \left(S_{2, j_1}^\dagger \right)^2 \\ & - \frac{2C_{S_2^1 S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_{2, j_1} y_{2, i_1, r_1} S_{2, j_1}^\dagger}{\sqrt{3}} \\ & - \frac{4}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} y_{2, i_2, r_2}} S_{2, j_1} S_{2, j_2} y_{2, i_1, r_2} y_{2, i_2, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger \\ & - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_{2, j_3} S_{2, j_4} y_{2, i_1, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger I_{S_2, j_3, j_1, j_4, j_2} \lambda_{S_2}^{(1)}}{\sqrt{3}} \\ & - \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} S_{2, j_3} S_{2, j_4} y_{2, i_1, r_1} S_{2, j_1}^\dagger S_{2, j_2}^\dagger I_{S_2, j_3, j_1, j_4, j_2} \lambda_{S_2}^{(2)}}{\sqrt{3}} \end{aligned}$	$-\frac{1}{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	
cont.										

Field combinations 6 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F W_{\mu\nu}$	$H \ell e$	$S_1 S_2 H$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$\begin{aligned} & + \frac{\overline{hy}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{3}{8} \frac{\overline{hy}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{\overline{hy}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{3}{8} \frac{\overline{hy}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{\overline{hy}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\ & + \frac{3}{8} \frac{\overline{hy}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\ & + \frac{\overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\ & + \frac{3}{8} \frac{\overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \end{aligned}$	$\begin{aligned} & - \frac{2\overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} \overline{H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} I_{y_{1,a_1,j_1} T_{y_{2,a_2,j_2}}} H_{a_1}^\dagger}}{\epsilon} \\ & - \frac{2\overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} \overline{H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} I_{y_{1,a_1,j_1} T_{y_{2,a_2,j_2}}} H_{a_1}^\dagger}}{\epsilon} \\ & \quad + \overline{\lambda T_3 S_{2,j_2} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & - \frac{\overline{hy}_{1,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{1,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & - \frac{1}{4} \frac{\overline{hy}_{1,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{1,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & - \frac{\overline{hy}_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{2,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & - \frac{1}{4} \frac{\overline{hy}_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{2,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & \quad + H_{a_1} H_{a_2} S_{1,j_2} T_{T_3,a_1,a_2,j_2,j_1} \lambda T_3 S_{2,j_1}^\dagger \\ & - \frac{\overline{hy}_{1,i_1,r_1} \overline{H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} T_{T_3,a_1,a_2,j_2,j_1} \lambda T_3 S_{2,j_1}^\dagger}}{2\epsilon} \\ & - \frac{1}{4} \frac{\overline{hy}_{1,i_1,r_1} \overline{H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} T_{T_3,a_1,a_2,j_2,j_1} \lambda T_3 S_{2,j_1}^\dagger}}{2\epsilon} \\ & - \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} T_{T_3,a_1,a_2,j_2,j_1} \lambda T_3 S_{2,j_1}^\dagger}}{2\epsilon} \\ & - \frac{1}{4} \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} T_{T_3,a_1,a_2,j_2,j_1} \lambda T_3 S_{2,j_1}^\dagger}}{2\epsilon} \\ & - \frac{2\overline{hy}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} \overline{H_{a_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} I_{y_{2,a_1,j_1} T_{y_{1,a_2,j_1}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{\epsilon} \\ & - \frac{2\overline{hy}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} \overline{H_{a_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} I_{y_{2,a_1,j_1} T_{y_{1,a_2,j_1}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{\epsilon} \\ & \quad + \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} T_{y_{1,a_1,j_1} T_{y_{2,a_2,j_2}}} H_{a_1}^\dagger \lambda_{H_2}^{(1)}}}{\epsilon} \\ & + \frac{1}{2} \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} T_{y_{1,a_1,j_1} T_{y_{2,a_2,j_2}}} H_{a_1}^\dagger \lambda_{H_2}^{(1)}}}{\epsilon} \\ & + \frac{\overline{hy}_{1,i_1,r_1} \overline{H_{a_2} y_{2,i_1,r_1} T_{y_{2,a_1,j_1} T_{y_{1,a_2,j_2}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H_2}^{(1)}}}{\epsilon} \\ & + \frac{1}{2} \frac{\overline{hy}_{1,i_1,r_1} \overline{H_{a_2} y_{2,i_1,r_1} T_{y_{2,a_1,j_1} T_{y_{1,a_2,j_2}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H_2}^{(1)}}}{\epsilon} \\ & + \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(cg1)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H_2}^{(2)}}}{\sqrt{3}\epsilon} \\ & + \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(cg1)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H_2}^{(2)}}}{2\sqrt{3}} \\ & - \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} T_{y_{1,a_1,j_1} T_{y_{2,a_2,j_2}}} H_{a_1}^\dagger \lambda_{H_2}^{(2)}}}{\sqrt{3}\epsilon} \\ & - \frac{\overline{hy}_{2,i_1,r_1} \overline{H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} T_{y_{1,a_1,j_1} T_{y_{2,a_2,j_2}}} H_{a_1}^\dagger \lambda_{H_2}^{(2)}}}{2\sqrt{3}} \\ & + \frac{\overline{hy}_{1,i_1,r_1} \overline{cg1_{j_1,j_2}} \overline{H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H_2}^{(2)}}}{\sqrt{3}\epsilon} \\ & + \frac{\overline{hy}_{1,i_1,r_1} \overline{cg1_{j_1,j_2}} \overline{H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H_2}^{(2)}}}{2\sqrt{3}} \\ & - \frac{\overline{hy}_{1,i_1,r_1} \overline{H_{a_2} y_{2,i_1,r_1} T_{y_{2,a_1,j_2} T_{y_{1,a_2,j_1}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H_2}^{(2)}}}{\sqrt{3}\epsilon} \\ & - \frac{\overline{hy}_{1,i_1,r_1} \overline{H_{a_2} y_{2,i_1,r_1} T_{y_{2,a_1,j_2} T_{y_{1,a_2,j_1}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H_2}^{(2)}}}{2\sqrt{3}} \end{aligned}$
cont.										
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A1} \right)^2$	$\begin{aligned} & + \frac{\sqrt{3} \overline{hy}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{3}{8} \frac{\sqrt{3} \overline{hy}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{\sqrt{3} \overline{hy}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{3}{8} \frac{\sqrt{3} \overline{hy}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\ & + \frac{\sqrt{3} \overline{hy}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\ & + \frac{3}{8} \frac{\sqrt{3} \overline{hy}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\ & + \frac{\sqrt{3} \overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\ & + \frac{3}{8} \frac{\sqrt{3} \overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \end{aligned}$	$\begin{aligned} & - \frac{2\overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} \overline{H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \text{Bar}(cg4)_{a_2,j_1,a_1,j_2} H_{a_1}^\dagger}}{\epsilon} \\ & - \frac{2\overline{hy}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} \overline{H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \text{Bar}(cg4)_{a_2,j_1,a_1,j_2} H_{a_1}^\dagger}}{\epsilon} \\ & \quad + \overline{\lambda T_3 S_{2,j_2} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & - \frac{\sqrt{3} \overline{hy}_{1,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{1,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & - \frac{1}{4} \frac{\sqrt{3} \overline{hy}_{1,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{1,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & - \frac{\sqrt{3} \overline{hy}_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{2,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & - \frac{1}{4} \frac{\sqrt{3} \overline{hy}_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_2} y_{2,i_1,r_1} T_{T_{3,a_1,a_2,j_1,j_2}}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \end{aligned}$
cont.										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F W_{\mu\nu}$	$H \ell e$	$S_1 S_2 H$
cont.									$+ \frac{\sqrt{3} h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+ \frac{3}{8} \sqrt{3} h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	$- \frac{1}{4} \sqrt{3} h \bar{y}_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,j_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$ $+ H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $- \frac{\sqrt{3} h \bar{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$
									$- \frac{1}{4} \sqrt{3} h \bar{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $- \frac{\sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\epsilon}$	$- \frac{1}{4} \sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $- \frac{2 h \bar{y}_{1,i_1,r_1} Y_{e,i_2,i_3} \text{cg}^4 a_{1,j_2,a_2,j_1} H_{a_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.									$- 2 h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3} \text{cg}^4 a_{1,j_2,a_2,j_1} H_{a_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $- \frac{h \bar{y}_{2,i_1,r_1} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}3)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}}{\epsilon}$	$- \frac{1}{2} h \bar{y}_{2,i_1,r_1} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}3)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$
									$- \frac{3 h \bar{y}_{2,i_1,r_1} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \text{Bar}(\text{cg}4)_{a_2,j_1,a_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}}{\epsilon}$	$- \frac{3}{2} h \bar{y}_{2,i_1,r_1} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \text{Bar}(\text{cg}4)_{a_2,j_1,a_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$ $- \frac{h \bar{y}_{1,i_1,r_1} \text{cg}3_{j_1,j_2} H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}}{\epsilon}$
cont.									$- \frac{1}{2} h \bar{y}_{1,i_1,r_1} \text{cg}3_{j_1,j_2} H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}$ $- \frac{3 h \bar{y}_{1,i_1,r_1} \text{cg}^4 a_{1,j_2,a_2,j_1} H_{a_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}}{\epsilon}$	$- \frac{3}{2} h \bar{y}_{1,i_1,r_1} \text{cg}^4 a_{1,j_2,a_2,j_1} H_{a_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}$
									$+ \frac{\sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}4)_{a_2,j_1,a_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\epsilon}$ $+ \frac{1}{2} \sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}4)_{a_2,j_1,a_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}$ $+ \frac{\sqrt{3} h \bar{y}_{1,i_1,r_1} \text{cg}^4 a_{1,j_1,a_2,j_2} H_{a_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}{\epsilon}$	$+ \frac{1}{2} \sqrt{3} h \bar{y}_{1,i_1,r_1} \text{cg}^4 a_{1,j_1,a_2,j_2} H_{a_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+ \frac{h \bar{y}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+ \frac{3}{8} h \bar{y}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+ \frac{\sqrt{3} h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+ \frac{3}{8} \sqrt{3} h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+ \overline{\lambda_{T3} S_{2,j_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}$ $- \frac{h \bar{y}_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{\epsilon}$
									$+ \frac{\sqrt{3} h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+ \frac{3}{8} \sqrt{3} h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+ \frac{h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+ \frac{3}{8} h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$	$- \frac{1}{2} h \bar{y}_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}$ $- \frac{h \bar{y}_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{\sqrt{3}\epsilon}$ $- \frac{h \bar{y}_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}}{2\sqrt{3}}$
cont.									$+ \frac{3}{8} h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ $+ \frac{\sqrt{3} h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+ \frac{3}{8} \sqrt{3} h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	$+ H_{a_1} H_{a_2} S_1 \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $- \frac{h \bar{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\epsilon}$ $- \frac{1}{2} h \bar{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$
									$+ \frac{3}{8} \sqrt{3} h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$ $+ \frac{\sqrt{3} h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+ \frac{3}{8} \sqrt{3} h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$	$- \frac{h \bar{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $- \frac{h \bar{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{2\sqrt{3}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+ \frac{\sqrt{3} h \bar{y}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+ \frac{3}{8} \sqrt{3} h \bar{y}_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+ \frac{h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+ \frac{3}{8} h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+ H_{a_1} H_{a_2} S_{1,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger$ $- \frac{h \bar{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\sqrt{3}\epsilon}$ $- \frac{h \bar{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{2\sqrt{3}}$
									$+ \frac{\sqrt{3} h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+ \frac{\sqrt{3} h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+ \frac{3}{8} \sqrt{3} h \bar{y}_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$	$- \frac{h \bar{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\epsilon}$ $- \frac{1}{2} h \bar{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger$ $+ \overline{\lambda_{T3} S_2 \mathcal{I}_{T3}^* a_{1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H \ell e$	$S_1 S_2 H$
cont.									$+ \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$- \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3}} S_2 y_{1,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
										$- \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3}} S_2 y_{1,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
										$- \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3}} S_2 y_{2,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
cont.									$- \frac{1}{2} \hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3}} S_2 y_{2,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+ \frac{\sqrt{3} \hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+ \overline{\lambda_{T3}} S_{2,j_2} \mathcal{T}_{T3_{a_1,a_2,j_1,j_2}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+ \frac{3}{8} \sqrt{3} \hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$- \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3}} S_{2,j_2} y_{1,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1,j_2}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+ \frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$- \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3}} S_{2,j_2} y_{1,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1,j_2}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
cont.									$+ \frac{3}{8} \sqrt{3} \hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$- \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3}} S_{2,j_2} y_{2,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1,j_2}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+ \frac{\sqrt{3} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$- \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3}} S_{2,j_2} y_{2,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_1,j_2}}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
									$+ \frac{3}{8} \sqrt{3} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$	$+ H_{a_1} H_{a_2} S_{1,j_2} \mathcal{T}_{T3_{a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger$
cont.									$+ \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$- \frac{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+ \frac{3}{8} \sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	$- \frac{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger}{2\sqrt{3}}$
									$+ \frac{3}{8} \sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	$- \frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
cont.									$- \frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{T}_{T3_{a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger}{2\sqrt{3}}$	

Field combinations 7 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+C_{S_1^1 S_2} s_2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_1,j_2}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{1,j_3} S_{2,j_2} S_{2,j_4} y_{1,i_2,r_1} y_{2,i_1,r_2} \text{Bar}(\text{cg}1)_{j_1,j_3} \text{Bar}(\text{cg}1)_{j_3,j_4}}$ $+2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_2} y_{1,i_2,r_1} \text{Bar}(\text{cg}1)_{j_2,j_3} S_{1,j_1}^\dagger}$ $+2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_2} y_{2,i_2,r_1} \text{Bar}(\text{cg}1)_{j_2,j_3} S_{2,j_1}^\dagger}$	$+ \frac{3h \overline{y_{1,i_2,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger g_2^2}{4c M_F} + \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} S_{1,j_3} y_{1,i_2,r_1} \lambda_{r_1} \overline{I_{y_{2,a_2,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger}}{\sqrt{3}}$ $+ \frac{h \overline{y_{1,i_2,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger g_2^2}{4c M_F} - 2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,r_2} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,r_3} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} H_{a_1}^\dagger S_{1,j_1}^\dagger}$ $+ \frac{3h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1} a_2 S_{1,j_1}^\dagger g_2^2}{4c M_F} + \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{I_{y_{2,a_2,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger S_{1,j_2}^\dagger}}{\sqrt{3}}$	
cont.								$+C_{S_1^1 S_2} s_2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_1,j_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$	$+ \frac{3h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger g_2^2}{4c M_F} + \frac{h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger g_2^2}{4c M_F}$ $+ \frac{h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_1,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger g_2^2}{4c M_F} + \frac{3h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T S_{1,j_3} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1} a_2 S_{2,j_1}^\dagger g_2^2}{4c M_F}$	$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I_{H_{1,a_1,a_2,j_1,j_2}}^1} \overline{I_{H_{1,a_1,a_2,j_1,j_2}}^1} \lambda_{H_1}^{(1)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I_{H_{1,a_1,a_2,j_1,j_2}}^2} \overline{I_{H_{1,a_1,a_2,j_1,j_2}}^2} \lambda_{H_1}^{(2)}}$
cont.								$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \text{cg}1_{j_1,j_3} S_{2,j_2} y_{2,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger}$ $+2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \text{cg}1_{j_1,j_3} S_{2,j_2} y_{2,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger}$	$+ \frac{3h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P L L_{a_1,i_1} g_2^2 S_{2,j_2} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F} + \frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P L L_{a_1,i_1} g_2^2 S_{2,j_2} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $+ \frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{M_F} + \frac{h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$	
cont.								$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \text{cg}1_{j_1,j_3} \text{cg}1_{j_2,j_4} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger S_{2,j_4}^\dagger}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^1 \lambda_{12}^{(1)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^1 \lambda_{12}^{(1)}}$	$- \frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $- \frac{3h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{8M_F}$ $- \frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{2c M_F}$ $- \frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4M_F}$	
cont.								$+ \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{\sqrt{3}}$ $+ \sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $- \sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+ \sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$	$- \frac{h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $- \frac{3h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{8M_F}$ $- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{c M_F}$	
cont.								$- \sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+ \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_2,j_3} S_{1,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger}}{\sqrt{3}} \lambda_{12}^{(2)}$	$- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{2M_F}$ $- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $- \frac{3h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{8M_F}$	
cont.								$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$	$- \frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_1} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $- \frac{3h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_1} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{8M_F}$ $- \frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_1} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{2c M_F}$ $- \frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_1} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4M_F}$	
cont.								$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_2,j_3} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{S_2}^{(1)}}$ $+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_1,j_3} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}}$	$+ \frac{3h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4c M_F}$ $+ \frac{h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{L_{a_1,i_1} C P R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{M_F}$	
cont.								$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$	$- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{2M_F}$ $- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $- \frac{3h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{8M_F}$	
cont.								$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_2,j_3} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{S_2}^{(1)}}$ $+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_1,j_3} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}}$	$+ \frac{3h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4c M_F}$ $+ \frac{h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{L_{a_1,i_1} C P R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{M_F}$	
cont.								$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$	$- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{2M_F}$ $- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $- \frac{3h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{8M_F}$	
cont.								$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_2,j_3} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{S_2}^{(1)}}$ $+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_1,j_3} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}}$	$+ \frac{3h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4c M_F}$ $+ \frac{h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{L_{a_1,i_1} C P R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{M_F}$	
cont.								$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$	$- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{2M_F}$ $- \frac{h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{4c M_F}$ $- \frac{3h \overline{y_{1,i_3,r_1} y_{2,i_1,r_2}} y_{2,i_2,r_2} L_{a_2,i_2}^T C P L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_2}}^\dagger} S_{1,j_1}^\dagger}{8M_F}$	
cont.								$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_2,j_3} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{S_2}^{(1)}}$ $+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_1,j_3} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}}$	$+ \frac{3h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4c M_F}$ $+ \frac{h \overline{L_{a_1,i_1}} C P R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{L_{a_1,i_1} C P R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_{1,a_1,j_1}}^\dagger} \overline{I_{y_{2,a_2,j_1}}^\dagger} S_{2,j_1}^\dagger}{M_F}$	
cont.								$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I_{12,j_2,j_3,j_1,j_4}}^2 \lambda_{12}^{(2)}}$	<	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	$S_1 S_2 \ell$	$S_1 F H$
cont.											$\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
											$\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_3,r_1}y_{2,i_3,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4cM_F}$	
											$\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
cont.											$+\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
											$+\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
											$+\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.											$+\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
											$-\frac{h\overline{y}_{1,i_2,r_1}y_{2,i_1,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}cM_F}$	
											$\frac{h\overline{y}_{1,i_2,r_1}y_{2,i_1,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
cont.											$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}cM_F}$	
											$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
											$-\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}cM_F}$	
cont.											$-\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
											$-\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}cM_F}$	
											$-\frac{h\overline{L}_{a_1,i_1}CPr\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0				$+\frac{C\epsilon_{j_1^2,j_2}^2}{h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}\frac{h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(\text{eg3})_{j_2,j_3}}-\frac{3h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}I_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger g_2^2}{8cM_F}$	$-2\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\lambda_{T3}\text{Bar}(\text{cg1})_{a_1,j_1,a_2,j_2}$
											$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}I_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger g_2^2}{8M_F}$	$-\frac{4h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
											$+\frac{10h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_1}\text{Bar}(\text{eg3})_{j_2,j_3}S_{2,j_1}^\dagger}{3\sqrt{3}}$	$-2\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\lambda_{T3}^{\text{cg1}}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger S_{1,j_2}^\dagger$
											$+\frac{10h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_1}\text{Bar}(\text{cg3})_{j_2,j_3}S_{2,j_1}^\dagger}{3\sqrt{3}}$	$+2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^{-1}$
cont.											$+\frac{C\epsilon_{j_1^2,j_2}^2}{h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}\frac{h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y}_{1,i_1,r_1}\text{cg3}_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$	$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^{-1}\lambda_{H1}^{(1)}$
											$+\frac{2}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^{-2}\lambda_{H1}^{(2)}$
											$-\frac{10}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
cont.											$+\frac{8}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$-\frac{10}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$+\frac{10h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}\text{cg3}_{j_1,j_2}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}}$	
cont.											$+\frac{10h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}\text{cg3}_{j_1,j_2}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger}{3\sqrt{3}}$	
											$-\frac{5}{9}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}\text{cg3}_{j_1,j_2}\text{cg3}_{j_2,j_3}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger S_{2,j_4}^\dagger$	
											$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_4}^{-1}\lambda_{12}^{(1)}$	
cont.											$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_4}^{-1}\lambda_{12}^{(1)}$	
											$+\frac{2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg3})_{j_2,j_3}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}}$	
											$+3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$	
											$-3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$	
cont.											$+3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$	
											$-3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$	
											$+\frac{2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\text{cg3}_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}}$	
cont.											$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_4}^{-2}\lambda_{12}^{(2)}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
								$-\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} 2\lambda_{12}^{(2)}$		
cont.								$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg } 3)_{j_2,j_3} S_{2,j_1}^\dagger \lambda_{S_2}^{(1)}$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8cMF}$	
								$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg } 3_{j_1,j_3}} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}$	$+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{16MF}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{y_{2,i_3,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{4cMF}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{y_{2,i_3,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8MF}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_Y^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8cMF}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_Y^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8MF}$	
cont.									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{2MF}$	
									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8cMF}$	
									$+\frac{9h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16MF}$	
cont.									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4cMF}$	
									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8MF}$	
									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8MF}$	
cont.									$+\frac{9h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16MF}$	
									$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8cMF}$	
									$+\frac{9h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16MF}$	
cont.									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8cMF}$	
									$+\frac{9h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16MF}$	
									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{y_{2,i_3,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8cMF}$	
cont.									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{y_{2,i_3,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16MF}$	
									$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{y_{2,i_3,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4cMF}$	
									$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8MF}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_2,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8cMF}$	
									$+\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_2,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{16MF}$	
cont.									$+\frac{3h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^Tg_Y^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8cMF}$	
									$-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^Tg_Y^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8MF}$	
									$-\frac{\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{2MF}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8cMF}$	
									$+\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{16MF}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{4cMF}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{4cMF}$	
									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8MF}$	
									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{2cMF}$	
									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8MF}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8MF}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8cMF}$	
									$+\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{16MF}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8cMF}$	
									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8cMF}$	
									$+\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{16MF}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{8cMF}$	
									$+\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T3_{a_1,a_2,j_1,j_2}^*S_{2,j_1}^\dagger}{16MF}$	
									$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8cMF}$	
cont.									$-\frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{16MF}$	
									$-\frac{3\sqrt{3}h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^Tg_Y^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8cMF}$	
									$-\frac{\sqrt{3}h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^Tg_Y^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8MF}$	
cont.									$-\frac{\sqrt{3}h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{2MF}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_5,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8cMF}$	

[illegible]

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	ℓ	S_1	F	H
										$-\frac{\overline{hL_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$			
cont.										$-\frac{7h\overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} \epsilon M_F}$			
										$-\frac{7h\overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$			
cont.										$-\frac{7h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} \epsilon M_F}$			
										$-\frac{7h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F \ell$	$S_2 F H$	$S_2 F \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\begin{aligned} & +\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T S_{1, j_1} S_{1, j_2} W_{1, i_1, r_1} Y_{e, i_1, i_3} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} g_2^{(2)}}{8 M_F} \\ & +\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_1, r_1}} \overline{Y_{1, i_2, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger g_2^{(2)}}{8 M_F} \\ & +\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T g_1^c S_{1, j_1} S_{1, j_2} Y_{1, i_1, r_1} W_{1, i_2, r_1} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2}}{8 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T S_{1, j_1} S_{1, j_2} W_{1, i_1, r_2} W_{1, i_3, r_2} W_{1, i_3, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2}}{2 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T S_{1, j_1} S_{1, j_2} W_{1, i_1, r_1} W_{1, i_2, r_2} W_{1, i_3, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2}}{4 M_F} \\ & -\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T S_{1, j_1} S_{1, j_2} W_{1, i_1, r_2} W_{1, i_3, r_2} W_{2, i_2, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2}}{4 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T S_{1, j_1} S_{1, j_2} W_{1, i_1, r_2} W_{1, i_3, r_1} W_{2, i_2, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2}}{M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_2, r_1}} \overline{W_{1, i_3, r_2}} \overline{W_{2, i_1, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} W_{1, i_3, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{4 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_2, r_1}} \overline{W_{1, i_3, r_2}} \overline{W_{2, i_1, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} W_{1, i_3, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{4 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_2, r_1}} \overline{W_{1, i_3, r_2}} \overline{W_{2, i_1, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} W_{1, i_3, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{2 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_1, r_1}} \overline{W_{1, i_3, r_1}} \overline{W_{2, i_2, r_2}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} W_{2, i_3, r_2} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{4 M_F} \\ & +\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_1, r_1}} \overline{W_{1, i_2, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} g_1^c \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{8 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T S_{1, j_1} S_{1, j_2} Y_{1, i_1, r_1} W_{1, i_2, r_1} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} \lambda_{S_1}^{(1)}}{2 M_F} \\ & -\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_1, r_1}} \overline{W_{1, i_2, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} \overline{T}_{W_1, a_1, j_1} \overline{T}_{W_1, a_2, j_2} S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger \lambda_{S_1}^{(1)}}{2 M_F} \end{aligned}$	$\begin{aligned} & -2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \frac{\overline{W_{2, i_1, r_1}} \overline{Y_{e, i_2, i_3}} H_{a_2} S_{2, j_2} Y_{2, i_2, r_1} Y_{e, i_1, i_3} \overline{T}_{H 2, a_1, j_2} \overline{T}_{H 2, a_2, j_2} H_{a_1}{}^\dagger S_{2, j_1}{}^\dagger}{\sqrt{3}} \\ & -h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \frac{\overline{W_{2, i_1, r_1}} H_{a_2} S_{2, j_2} W_{2, i_1, r_1} H_{a_1}{}^\dagger S_{2, j_1}{}^\dagger \overline{T}_{H 2, a_1, j_2} \overline{T}_{H 2, a_2, j_2} \lambda_{H 2}^{(1)}}{M_F^2} \\ & -h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \frac{\overline{W_{2, i_1, r_1}} H_{a_2} S_{2, j_2} W_{2, i_1, r_1} H_{a_1}{}^\dagger S_{2, j_1}{}^\dagger \overline{T}_{H 2, a_1, j_2} \overline{T}_{H 2, a_2, j_2} \lambda_{H 2}^{(2)}}{M_F^2} \end{aligned}$	$\begin{aligned} & +\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{2, i_1, r_1}} \overline{Y_{2, i_2, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} S_{2, j_1} S_{2, j_2} \overline{T}_{W 2, a_1, j_1} \overline{T}_{W 2, a_2, j_2} g_2^{(2)}}{8 M_F} \\ & +\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T W_{2, i_1, r_1} Y_{2, i_2, r_1} \overline{T}_{W 2, a_1, j_1} \overline{T}_{W 2, a_2, j_2} S_{2, j_1}{}^\dagger S_{2, j_2}{}^\dagger g_2^{(2)}}{8 M_F} \\ & +\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{2, i_1, r_1}} \overline{Y_{2, i_2, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} g_1^c S_{2, j_1} S_{2, j_2} \overline{T}_{W 2, a_1, j_1} \overline{T}_{W 2, a_2, j_2}}{8 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{1, i_1, r_1}} \overline{Y_{2, i_2, r_1}} \overline{W_{2, i_3, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} S_{2, j_1} S_{2, j_2} W_{1, i_3, r_1} \overline{T}_{W 2, a_1, j_1} \overline{T}_{W 2, a_2, j_2}}{4 M_F} \\ & +\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{2, i_1, r_1}} \overline{Y_{2, i_2, r_1}} \overline{W_{2, i_3, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} S_{2, j_1} S_{2, j_2} W_{1, i_3, r_1} \overline{T}_{W 2, a_1, j_1} \overline{T}_{W 2, a_2, j_2}}{4 M_F} \\ & -\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{W_{2, i_1, r_1}} \overline{Y_{2, i_2, r_1}} \overline{W_{2, i_3, r_1}} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} S_{2, j_1} S_{2, j_2} W_{1, i_3, r_1} \overline{T}_{W 2, a_1, j_1} \overline{T}_{W 2, a_2, j_2}}{2 M_F} \\ & -\frac{3 h \log \left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}{}^T$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	ℓ	S_2					F	H	S_2					F	ℓ	
											$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,\tau_1}^1Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,\tau_1}Y_{e,i_3,i_3}H_{a_1}^{\dagger}S_{2,j_1}^{\dagger}}{\sqrt{3}}$														
T3-E	0	3	0	3		1	2		$\frac{1}{2}$		$+2i\sqrt{\frac{2}{3}}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,\tau_1}^1Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,\tau_1}Y_{e,i_3,i_3}H_{a_1}^{\dagger}S_{2,j_1}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_2,j_1}^{\dagger}$														
											$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,\tau_1}^1H_{a_2}S_{2,j_2}y_{2,i_1,\tau_1}}H_{a_1}^{\dagger}S_{2,j_1}^{\dagger}\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^{\dagger}2\lambda_{H2}^{(2)}}{\sqrt{3}}$														

Field combinations 9 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H	ℓ	S_1	S_2	F	H	ℓ	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1 y_{2, i_2}, \tau_1^*} L_{a_4, i_2}{}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda \tau_3 \epsilon_{a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} M_F} + \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{\lambda \tau_3 L_{a_3, i_1} C P_R L_{a_4, i_2}{}^T} y_{1, i_1}, \tau_1 y_{2, i_2}, \tau_1^* \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1^*} Y_{e, i_2, i_3} H_{a_2} S_{1, j_1} S_{2, j_2} y_{1, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_1} H_{a_1}^\dagger + \frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_2}, \tau_1 y_{2, i_1}, \tau_1^*} L_{a_2, i_2}{}^T C P_L L_{a_1, i_1} S_{2, j_2} \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_1} S_{1, j_1}^\dagger g_2^2}{4M_F} + \frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1 y_{2, i_2}, \tau_1^*} L_{a_4, i_2}{}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger g_2^2}{4M_F}$						
								$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} \overline{\lambda \tau_3} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{T3 a_1, a_2}, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger - \frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} \overline{\lambda \tau_3} S_{2, j_2} y_{2, i_1}, \tau_1 \overline{\tau_{T3 a_1, a_2}, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger$							$+ \frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1, i_1} C P_R L_{a_2, i_2}{}^T} S_{1, j_2} y_{1, i_2}, \tau_1 y_{2, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_2} \overline{\tau_{y_2, a_2}, j_2} S_{2, j_1}^\dagger g_2^2}{4M_F} + \frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1, i_1} C P_R L_{a_2, i_2}{}^T} S_{1, j_1} y_{1, i_1}, \tau_1 y_{2, i_2}, \tau_1 \epsilon_{a_1, a_2} S_{2, j_1}^\dagger g_2^2}{4M_F}$						
								$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_1} H_{a_2} S_{1, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{T3 a_1, a_2}, j_2} \lambda \tau_3 S_{2, j_1}^\dagger - \frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} H_{a_2} S_{1, j_2} y_{2, i_1}, \tau_1 \overline{\tau_{T3 a_1, a_2}, j_2} \lambda \tau_3 S_{2, j_1}^\dagger$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} Y_{e, i_2, i_3} H_{a_2} y_{2, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_2, a_1}, j_2} \overline{\tau_{y_1, a_2}, j_1} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger + h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$						
								$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
cont.																					
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								$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} Y_{e, i_2, i_3} H_{a_2} y_{2, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_2, a_1}, j_2} \overline{\tau_{y_1, a_2}, j_1} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger + h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
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cont.								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} \text{cg} 1_{j_1, j_2} H_{a_1} y_{2, i_1}, \tau_1 H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} Y_{e, i_2, i_3} H_{a_2} y_{2, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_2, a_1}, j_2} \overline{\tau_{y_1, a_2}, j_1} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger + h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} Y_{e, i_2, i_3} H_{a_2} y_{2, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_2, a_1}, j_2} \overline{\tau_{y_1, a_2}, j_1} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger + h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
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								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} Y_{e, i_2, i_3} H_{a_2} y_{2, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_2, a_1}, j_2} \overline{\tau_{y_1, a_2}, j_1} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger + h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
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cont.								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} \text{cg} 1_{j_1, j_2} H_{a_1} y_{2, i_1}, \tau_1 H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} Y_{e, i_2, i_3} H_{a_2} y_{2, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_2, a_1}, j_2} \overline{\tau_{y_1, a_2}, j_1} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger + h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$							$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} Y_{e, i_2, i_3} H_{a_2} y_{2, i_2}, \tau_1 Y_{e, i_1, i_4} \overline{\tau_{y_2, a_1}, j_2} \overline{\tau_{y_1, a_2}, j_1} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger + h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$						
								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_2} S_{1, j_2} S_{2, j_1} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$							$+ h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} H_{a_2} y_{2, i_1}, \tau_1 \overline{\tau_{y_2, a_1}, j_1} \overline{\tau_{y_1, a_2}, j_2} H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(1)} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \text{Bar}(\text{cg} 1)_{j_1, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \tau_1} H_{a_1} S_{1, j_1} S_{2, j_2} y_{1, i_1}, \tau_1 \overline{\tau_{y_1, a_1}, j_1} \overline{\tau_{y_2, a_2}, j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$						
cont.																					
cont.								$+ \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1}, \tau_1} \text{cg} 1_{j_1, j_2} H_{a_1} y_{2, i_1}, \tau_1 H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}} + \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1}, \$													

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_1 y_{1,i_2,r_1} i_3 y_{1,i_3,r_2} y_{2,i_1,r_2} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$	$\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_1 y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$
									$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{M_F}$	$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{\sqrt{3} M_F}$
									$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_1 y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_1} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$	$\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$
cont.									$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}^{-T} C P_L L_{a_1,i_1} S_2 S_{j_1} T r_{3,a_1,a_2,j_1}^* S_1^\dagger \lambda_{12}^{(1)}}{M_F}$	$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^{-T} S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} T r_{3,a_1,a_2,j_1}^* S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_4,i_2}^{-T} C P_L L_{a_2,i_1} H_{a_3} \lambda_{T3} e_{a_1,a_2} e_{a_3,a_4}}{\sqrt{3} M_F}$ $+ \frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R L_{a_4,i_2}^{-T} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{a_1,a_3} e_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} T r_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\sqrt{3}}$ $- h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} T r_{3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\sqrt{3}}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \lambda_{T3} S_2 y_{1,i_1,r_1} T r_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_1 j_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$ $\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_{j_1} j_1 y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$ $\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_{j_1} j_1 y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_1} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{\sqrt{3} M_F}$
cont.									$- h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \lambda_{T3} S_2 y_{2,i_1,r_1} T r_{3,a_1,a_2,j_1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_{j_1} j_1 y_{1,i_2,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$
									$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_{j_1} j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{M_F}$	$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^{-T} S_{j_1} j_1 y_{1,i_3,r_2} y_{2,i_2,r_1} T r_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4 M_F}$
									$\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^{-T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} T r_{3,a_1,a_2,j_1}^* S_{1,j_1}^\dagger}{4 M_F}$	$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^{-T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} T r_{3,a_1,a_2,j_1}^* S_{1,j_1}^\dagger}{\sqrt{3} M_F}$
cont.									$\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^{-T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} T r_{3,a_1,a_2,j_1}^* S_{1,j_1}^\dagger}{4 M_F}$	$\frac{h \log \left(\frac{\$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	S_1S_2FH	$S_1S_2F\ell$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$ $-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_4,r_2} I \tau \tilde{g}_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T g_{S_{1,j_2}^*}^* y_{1,i_1,r_1} y_{2,i_2,r_1} I \tau \tilde{g}_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger}{M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_3} y_{1,i_3,r_2} y_{2,i_4,r_1} I \tau \tilde{g}_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
cont.										$-\frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_4,r_2} I \tau \tilde{g}_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger}{3M_F}$ $-\frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} I \tau \tilde{g}_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger}{3M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} I \tau \tilde{g}_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I \tau \tilde{g}_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
cont.										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_3} y_{2,i_4,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$ $-\frac{2\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T g_{S_{1,j_1}}^* y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{M_F}$
										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_4,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3M_F}$ $-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+\frac{\sqrt{\frac{2}{3}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} I \tau \tilde{g}_{a_1,a_2,j_2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$ $+\frac{\sqrt{\frac{2}{3}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} I \tau \tilde{g}_{a_1,a_2,j_2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$-\frac{3\sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} I \tau \tilde{g}_{a_1,a_2,j_2,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$ $-\frac{3\sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I \tau \tilde{g}_{a_1,a_2,j_2,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$
										$-\frac{7h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$
cont.										$-\frac{7h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$

Field combinations 10 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell e$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
cont.								
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
cont.								
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
cont.								
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
cont.								
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
cont.								